

SUMMER TRAINING REPORT

RSWM LIMITED

A Project Report Submitted by

JEEVESH

OF

Computer Science and Engineering

IILM University, Greater Noida

Batch: 2024 – 2028

Under supervision of

Mr. Anil Yadav

(Industry Mentor, RSWM Limited)

Duration: 8 June 2026 – 3 July 2026

Date of Submission: 2 July 2026

DECLARATION

I, JEEVESH, student of B.Tech (Computer Science and Engineering), Batch 2024–2028, declare that the SUMMER TRAINING REPORT ON CYBERSECURITY which is submitted by me to the Department of Computer Science and Engineering, IILM University, Greater Noida, under the guidance of Mr. Anil Yadav (Industry Mentor, RSWM Limited), is a record of my own work carried out during the internship period from 8 June 2026 to 3 July 2026.

I further declare that this report has not been submitted elsewhere for the award of any degree or diploma. The information presented in this report is true and original to the best of my knowledge.

Date: 2 July 2026

JEEVESH

B.Tech CSE (2nd Year)

IILM University

ACKNOWLEDGEMENT

It is with profound gratitude that I express my deep indebtedness to all the employees of RSWM Limited, who extended their wholehearted support and guidance throughout my internship. Without their cooperation, it would not have been possible for this training to have materialized and taken a concrete shape.

I owe my sincere thanks to Mr. Anil Yadav (Industry Mentor, RSWM Limited) for his invaluable guidance, encouragement, and constant support during the entire training period. His expertise in cybersecurity helped me gain practical knowledge and hands-on experience.

I would also like to thank the faculty and administration of IILM University, Greater Noida, for providing me with this opportunity to undertake summer training as part of my B.Tech curriculum.

Last but not least, I am grateful to my family and friends for their constant motivation and support throughout this journey.

JEEVESH
B.Tech CSE (2nd Year)
IILM University

PREFACE

Organizations are made up of people and function through people. Without people, organizations cannot exist. The resources of men, money, material, machinery, and mechanism are connected, coordinated, and utilized through people. Engineers need to concentrate more on mechanism and the way in which things have been made. The need of training arises for doing things yourself, understanding its way.

Practical exposure for doing things makes a person conversant to the technicalities involved in any job. In view of such benefits, imparting vocational training has been made an integral part of any academic structure.

At RSWM Limited, training is given to Engineering Aspirants to secure their future in the dynamic world of Cybersecurity. Today, this industry is one of the very fastest growing fields in the world, with organizations increasingly relying on skilled professionals to protect their digital assets and infrastructure.

In this regard, I have undertaken 4 weeks of summer training at RSWM Limited. In my report, I try to introduce the concepts of Cybersecurity and its practical implementation and setup at RSWM Limited. This training has helped me bridge the gap between theoretical knowledge gained at IILM University and its real-world application in the industry.

JEEVESH

B.Tech CSE (2nd Year)

IILM University

HISTORY OF LNJ BHILWARA GROUP

About LNJ Bhilwara Group

The incredible journey of LNJ Bhilwara Group started in 1960 when founder Mr. L. N. Jhunjhunwala set up a modest textile mill in Bhilwara, Rajasthan. Over the decades, with robust vision, diligence and unwavering commitment, they grew into one of the country's leading corporate groups with businesses in textiles, graphite electrodes, hydro power generation, energy storage solutions and IT services.

The Group today stands proud as a multi-products and services conglomerate with an annual turnover in excess of INR 9,691 Crore (\$1.2 Billion). The Group's 8 Companies are IS/ISO Certified and 5 Companies are listed with stock exchanges in India.

Group Companies

- RSWM Limited
- HEG Limited
- Maral Overseas Limited
- Bhilwara Info Technology Limited
- Bhilwara Technical Textiles Limited
- Bhilwara Energy Limited (Power)

RSWM Limited

- One of India's largest yarn manufacturers
- Chairman & M.D. – Sh. Riju Jhunjhunwala
- C.I.O – Sh. Naresh Sharma
- Turnover – INR 4000 Crore
- Provides the best quality of yarns and fabrics
- Headquarters: Bhilwara, Rajasthan, India

RSWM Limited – Detailed Overview

RSWM Limited, formerly known as Rajasthan Spinning & Weaving Mills Limited, is a flagship company of the LNJ Bhilwara Group. Established in 1960, RSWM is one of India's leading integrated textile companies, engaged in the manufacturing of high-quality yarn and fabric. The company has a strong domestic and international presence, exporting its products to over 50 countries worldwide.

RSWM operates multiple state-of-the-art manufacturing units across Rajasthan, including plants at Bhilwara, Kharigram, Gulabpura, and Ringas. The company produces a wide range of yarn varieties including polyester, viscose, cotton, and blended yarns, catering to diverse customer requirements across domestic and global markets.

Key Facts

- Founded: 1960
- Industry: Textile – Yarn & Fabric Manufacturing
- Listed on: Bombay Stock Exchange (BSE) & National Stock Exchange (NSE)
- Annual Turnover: INR 4000+ Crore

- Employees: 20,000+
- Export Markets: 50+ countries across Europe, USA, and Asia
- Certifications: ISO 9001, ISO 14001, OEKO-TEX Standard 100

Cybersecurity at RSWM Limited

As a large-scale manufacturing enterprise with significant digital infrastructure, RSWM Limited places great importance on cybersecurity. The company manages an extensive IT network spanning multiple plants and offices, making the protection of data, systems, and communication channels critical to its operations.

The IT and cybersecurity team at RSWM is responsible for maintaining network security, implementing firewalls and intrusion detection systems, managing user access controls, and ensuring compliance with data protection standards. The company regularly conducts security audits and employee awareness programs to strengthen its cybersecurity posture.

During my 4-week internship at RSWM Limited, I had the opportunity to work alongside the cybersecurity team and gain hands-on exposure to real-world security practices followed in a large industrial organization.

IT INFRASTRUCTURE

What Is IT Infrastructure?

Today, every company is a technology company. This has proven true throughout the era of digital transformation. No matter what industry you are in, what you are selling, or who you are selling to, technology plays a vital role in how you make, deliver, and market your products and services.

As a result, IT infrastructure is no longer a set-it-and-forget-it support system underneath more important business processes. Now, IT infrastructure management is a main driver of business success. If not handled properly, your business risks falling behind more agile and efficient competitors.

On a high level, IT infrastructure is the backbone of systems and processes that allow an organization to deliver solutions and services to the workforce and customers. More specifically, IT infrastructure management includes maintenance for:

Hardware: The servers, data centers, switches, and other on-premises equipment that support IT services.

Cloud Computing: IT infrastructure management is not just about on-premises equipment anymore. It must also cover the various use cases for public, private, and hybrid cloud technology.

Software: Various applications and services that are deployed throughout the organization — CRM, ERP, Communications apps, etc.

Networking: LAN and WAN connectivity as well as cybersecurity appliances like firewalls.

Human Processes: All of the administrators, developers, and end users with access to IT systems and the processes required to maintain those systems and services.

Four IT Infrastructure Management Challenges

Availability: Your workforce depends on the uptime of mission-critical systems and applications. One misstep in IT infrastructure management can result in downtime that costs the business millions of Rupees in lost productivity. Strategically planning for business continuity is essential.

Budgeting: With so many varying economic models within the IT infrastructure, financial planning can be challenging. When tech leaders are asked to do more with less, IT infrastructure training must include the financial side of the conversation.

Agility: Speed is everything for modern IT infrastructure management. If the IT department is not prepared to adapt quickly to new business demands, you risk falling behind competitors.

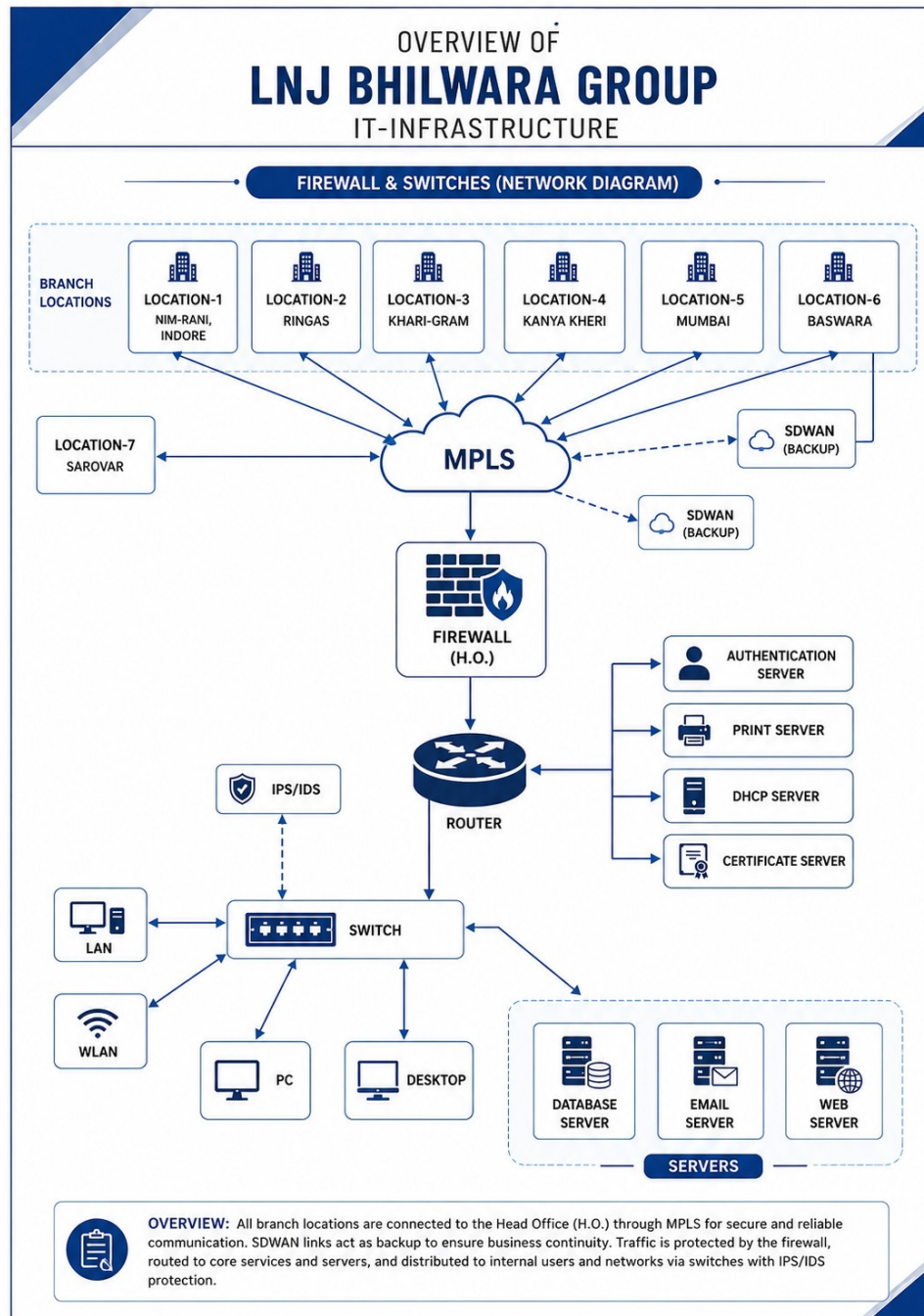
Control: The modern IT infrastructure is not just on-premises. Moving to the cloud helps with agility and budgeting but takes some control out of IT's hands. IT infrastructure training must teach leaders to control systems even when they are not fully owned.

Internet

The Internet is a global network that connects billions of computers across the world with each other and to the World Wide Web. It uses the standard internet protocol suite (TCP/IP) to connect billions of computer users worldwide. The Internet carries a vast range of information resources and services, such as the inter-linked hypertext documents and applications of the World Wide Web, electronic mail, telephony, and file sharing.

LNJ BHILWARA GROUP — IT INFRASTRUCTURE OVERVIEW

This overview explains every component of LNJ Bhilwara Group's IT network, from branch connectivity through to internal services and end-user access, as shown in the network diagram.



Branch Locations

Seven branch sites — Nim-Rani Indore, Ringas, Khari-Gram, Kanya Kheri, Mumbai, Baswara, and Sarovar — are each connected back to the Head Office. These are the company's operational locations that need continuous access to centralized systems and data.

MPLS (Primary WAN Link)

MPLS is the main network technology that links all branch locations to the Head Office. It provides a reliable, low-latency private connection, allowing all sites to communicate securely and consistently as if on one unified network.

SDWAN (Backup Links)

Two SDWAN connections sit alongside MPLS as backup paths. If the primary MPLS link ever fails or becomes congested, traffic automatically reroutes through SDWAN, keeping branch offices connected to the Head Office without interruption.

Firewall (Head Office)

All traffic arriving from MPLS/SDWAN passes through the Head Office firewall first. It inspects and filters this traffic, blocking unauthorized or malicious activity before anything reaches internal systems.

Router

After the firewall, the router acts as the central traffic director. It sends data to the four core services (authentication, print, DHCP, certificate) on one side, and to the internal switch on the other.

IPS/IDS

Connected to the switch, the Intrusion Prevention/Detection System continuously watches internal network traffic for suspicious behavior, adding a second layer of protection behind the firewall.

Switch

The switch is the hub for everything inside the Head Office network — it connects the router, IPS/IDS, end-user devices (LAN/WLAN, PCs, desktops), and the server farm, allowing all of them to communicate with each other.

Core Network Services

Service	Function
Authentication Server	Verifies user logins and controls network access
Print Server	Manages print jobs across the organization
DHCP Server	Assigns IP addresses to connecting devices
Certificate Server	Issues digital certificates for secure communication

Server Farm

Database, Email, and Web servers sit behind the switch, storing company data, handling email communication, and hosting web applications respectively.

End-User Access (LAN / WLAN / PC / Desktop)

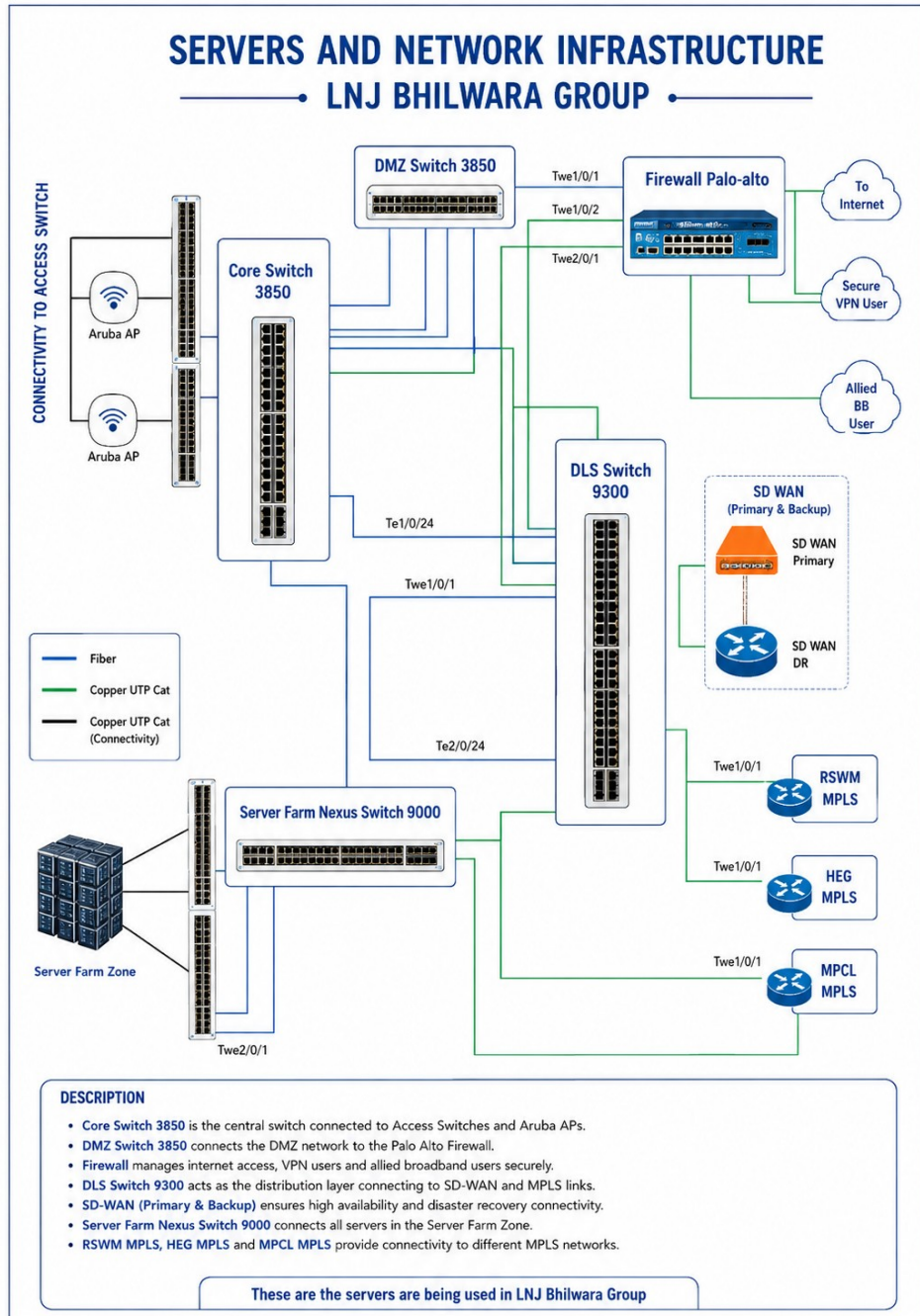
Employees connect to the network through wired LAN or wireless WLAN, using PCs and desktops, all linked through the switch — giving them access to core services, the internet, and internal servers.

Conclusion

Together, these components form a secure, centralized network: branches reach the Head Office via MPLS with SDWAN backup, traffic is firewall-filtered, routed to core services and servers, and made available to end users through a switch protected by IPS/IDS.

SERVICES & NETWORK INFRASTRUCTURE — LNJ BHILWARA GROUP

This overview explains every element of the servers and network infrastructure diagram, covering access connectivity, core and distribution switches, the firewall, SD-WAN, MPLS links, and the server farm.



Access Layer (Aruba APs)

Two Aruba access points provide wireless connectivity for end users. They connect through access switches into the Core Switch 3850 using copper UTP cabling.

Core Switch 3850

This is the central switch of the network. It connects to the access switches/Aruba APs, the DMZ Switch 3850, the DLS Switch 9300, and the Server Farm Nexus Switch 9000, tying the entire infrastructure together.

DMZ Switch 3850 & Firewall (Palo Alto)

The DMZ Switch connects to the Palo Alto firewall over fiber uplinks (Twe1/0/1, Twe1/0/2, Twe2/0/1). The firewall secures and manages three types of external traffic: general internet access, secure VPN users, and Allied BB (broadband) users.

DLS Switch 9300

Acts as the distribution layer switch, sitting between the Core Switch and external connectivity. It links onward to SD-WAN and the three MPLS providers.

SD-WAN (Primary & Backup)

A primary SD-WAN appliance and a backup/DR router provide redundant WAN connectivity, ensuring high availability and disaster recovery if the primary path fails.

MPLS Connections

Three separate MPLS links — RSWM MPLS, HEG MPLS, and MPCL MPLS — connect through the DLS Switch 9300, linking to other group companies or business units within LNJ Bhilwara Group.

Server Farm Nexus Switch 9000

Connects directly to the physical Server Farm Zone, linking all servers into the core network via fiber connections.

Cabling Legend

Cable Type	Color	Used For
Fiber	Blue	High-speed links between core, distribution, and server farm switches
Copper UTP Cat	Green	Firewall, DMZ, SD-WAN, and MPLS connections
Copper UTP Cat (Connectivity)	Black	Access switch to Aruba AP connectivity

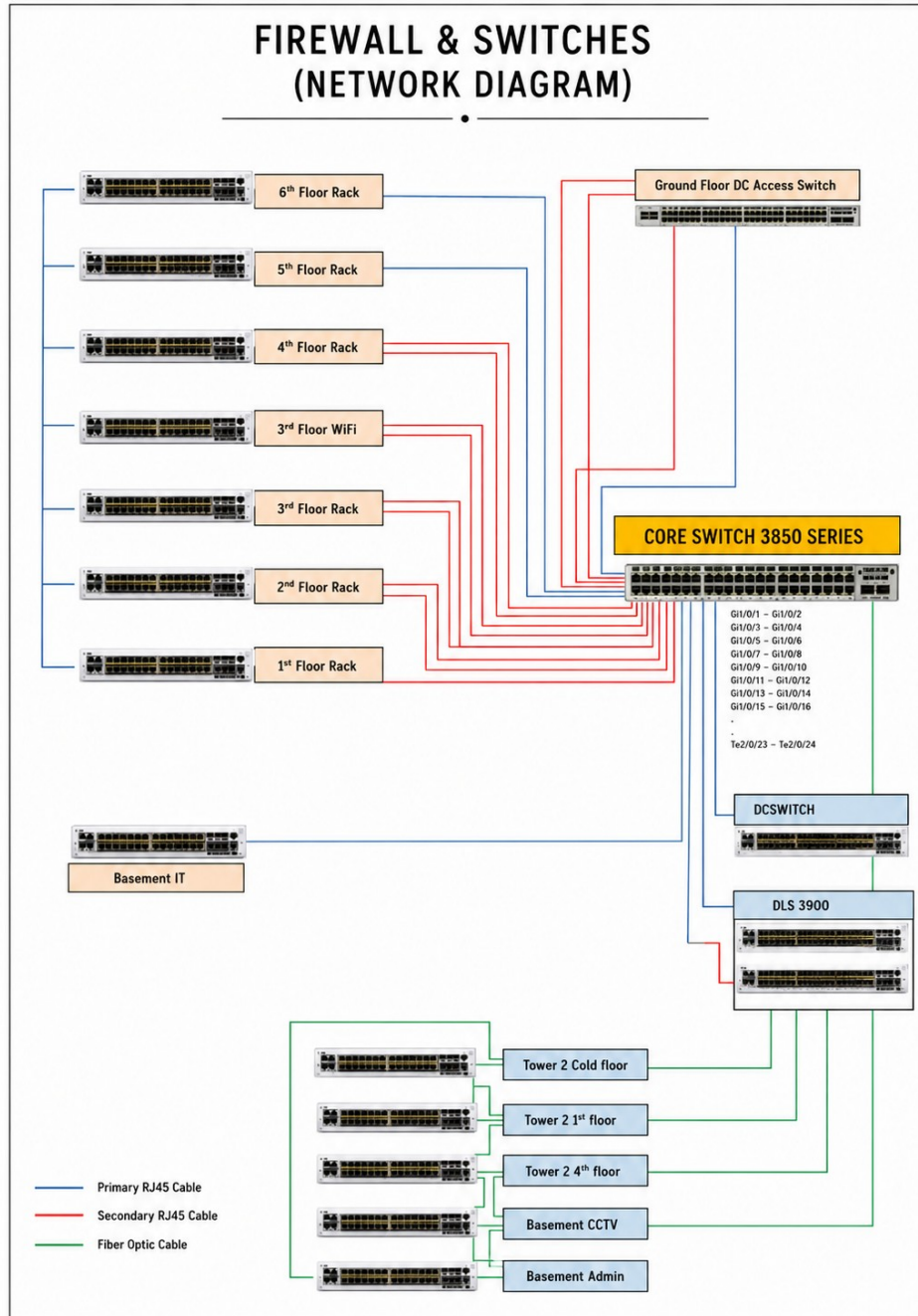
Conclusion

The infrastructure follows a layered design: wireless access points and access switches feed into the Core Switch 3850, which connects to the DMZ/firewall for secure internet, VPN, and broadband access, and to the DLS Switch 9300 for SD-WAN and MPLS connectivity to other

group entities. The Server Farm Nexus Switch 9000 links all servers into this same core network, forming the complete infrastructure used across LNJ Bhilwara Group.

FIREWALL & SWITCHES — NETWORK DIAGRAM OVERVIEW

This overview explains every element of the switching network diagram, covering the floor-wise rack switches, the core switch, distribution switches, Tower 2 connectivity, and the cabling types used to link them.



Tower 1 Floor Rack Switches

Seven access switches serve Tower 1 — covering the 1st through 6th floor racks plus a dedicated 3rd Floor WiFi switch. Each connects directly into the Core Switch using RJ45 cabling, giving every floor its own switch for local device connectivity (PCs, APs, and other endpoints).

Core Switch (3850 Series)

The Cisco Catalyst 3850 Series Core Switch is the central aggregation point of the network. All floor rack switches, the Ground Floor DC Access Switch, the DCSWITCH, and the DLS 3900 connect into it. It uses defined port ranges (Gi1/0/1 through Gi1/0/16, and Te2/0/23-24 for higher-speed uplinks) to organize connections by device.

Ground Floor DC Access Switch

This switch connects the Ground Floor data center area to the Core Switch, providing network access for ground-floor data center equipment.

Basement IT

A separate switch for the Basement IT area connects directly to the floor rack switch chain via a primary RJ45 link, extending network access to the basement IT room.

DCSWITCH & DLS 3900

Downstream of the Core Switch, the DCSWITCH and the DLS 3900 (a dual-switch stack) form the next distribution layer, extending connectivity further into the data center and toward Tower 2 via fiber.

Tower 2 Connectivity

Five switches serve Tower 2 locations — Cold Floor, 1st Floor, 4th Floor, Basement CCTV, and Basement Admin. These connect back to the DLS 3900 using fiber optic cabling, suited for longer-distance, high-bandwidth links between towers.

Cabling Legend

Cable Type	Color	Used For
Primary RJ45 Cable	Blue	Main floor switches (5th, 6th floor, Basement IT) to Core Switch
Secondary RJ45 Cable	Red	Remaining floor switches (1st-4th floor, 3rd floor WiFi) to Core Switch
Fiber Optic Cable	Green	Tower 2 switches to DLS 3900 distribution switch

Conclusion

The network follows a layered switching design: floor-level access switches in Tower 1 connect into a central Core Switch (3850 Series) using RJ45 cabling, which in turn feeds the DCSWITCH and DLS 3900 distribution layer. From there, fiber optic links extend connectivity to Tower 2's floor and basement switches, creating a structured, scalable campus network.

DATA CENTRE MONITORING SYSTEM

Monitoring

Monitoring is the process of observing, measuring and recording the performance or behaviour of a system, process, or activity over time.

Dashboard Monitoring: Shift Manager monitors all the application and equipment inside Data Centre by using a Dashboard.



The Dashboard provides quick navigation to every monitored system in the Data Centre, including PAC-1, PAC-2, PAC-3, Purafil, UPS-1, UPS-2, Battery Monitoring, the M3 grid views (Production, Test, and Development), Network, DC Camera Monitoring, BI Server, HMC for I Series and DR I Series Servers, the RSWM/MOL/DR Tape Libraries, the M3 Server for QBatch and MOVEX Mail, and the JBOSS login screen on the NOW Server.

PAC-1 (PeX) Status

The PeX unit, located in the Server Room of the HO Datacenter, is shown below in Normal Operation with both unit and communications status green. The panel reports live Return Air temperature and humidity along with fan speed, cooling, free cooling, next maintenance due date, reheating, dehumidifying, and humidifying levels.



PAC Units & Infrastructure Health

PAC units (PeX, PDX-PCW), powered by Liebert iCOM, control temperature and humidity in the server room, tracking Return Air / Supply Air readings, fan speed, cooling, and maintenance status. The Infrastructure Health panel summarizes the up/down status of all monitored devices (Windows, UPS, Firewall, Routers, Switches) and active alerts.

PDX-PCW, Purafil & Infrastructure Health

The PDX-PCW unit displays standby status, return/supply air temperature, and humidity. The Purafil OnGuard 4000 system tracks corrosion levels (copper and silver sensors) along with temperature and humidity, flagging alarms if thresholds are breached. A second Infrastructure Health snapshot reflects the live device status across the network.

UPS, Infrastructure Health & CCTV

The UPS dashboard (Emerson Liebert HiPulse) shows Bypass, Input, Battery, and Output stages with live voltage, amperage, and load data, plus active alarms such as a bypass switch being open. The CP Plus CCTV system provides live camera feeds from the Network Room, Datacentre Front/Back, Fire Control, and UPS Rooms, with PTZ camera controls.

DATA CENTRE AND NETWORK SECURITY POLICY

Network Access Controls

1. **Firewall Routing:** Route all LAN connections through a firewall to prevent unauthorized access.
 - Review firewall rule sets at least quarterly to remove outdated or unused rules.
2. **Secure Design:** Securely design and manage the network to protect against internal and external threats.
 - Conduct periodic vulnerability assessments and penetration testing of the network.
3. **Network Segregation:** Segregate external and internal IT systems, and monitor network devices for outages.
 - Implement VLANs to separate guest, corporate, and critical infrastructure traffic.
4. **Password Management:** They change default passwords on network devices.
 - Enforce strong password complexity and periodic password rotation for all network devices.
5. **Role-Based Access:** They Allocate network access based on user roles.
 - Review user access rights at least every six months and revoke access promptly upon role change or exit.
6. **Admin Rights:** They do not grant superuser or network admin rights for daily tasks.
 - Maintain a separate, logged admin account used only for administrative activities.
7. **Remote Access:** Granted remote access only for authorized business purposes with departmental approval.
 - Enforce multi-factor authentication (MFA) for all remote access connections.
8. **Inventory and Documentation:** Maintain network device inventory and document network topology for management and troubleshooting.
 - Update network diagrams and asset inventory whenever new devices are added or decommissioned.

Physical Access Controls

9. **Secure Environment:** Core network equipment and servers are in controlled, secure environments.
 - Restrict environmental control settings to authorized IT personnel only.
10. **Data Centre/Server Room:** Corporate business-critical applications are in the Data Centre; remote locations have Server Rooms.
 - Maintain an updated list of all critical applications and their hosting locations.
11. **Access Restriction:** Only authorized IT staff can access the Data Centre/Server Room, with the user list reviewed periodically.
 - Use biometric or card-based access control with individual access logs.
12. **Visitor Access:** Visitor access requires a legitimate business purpose and approval, with entry/exit details logged.
 - Visitors must be escorted by authorized personnel at all times within secure areas.
13. **Environmental Controls:** Critical servers must have monitored temperature, backup power, and fire suppression systems.
 - Set up automated alerts for temperature, humidity, or power threshold breaches.
14. **CCTV Monitoring:** Use CCTV for security, retaining recordings for at least 30 days, and consider voice recording and motion sensors for new solutions.

- Position cameras to cover all entry/exit points and server racks without blind spots.

UPS (Uninterruptible Power Supply) Equipment Security Controls

- 15. UPS Provision:** Ensure UPS for all IT equipment to maintain operations during power outages.
 - Conduct periodic load testing of UPS systems to confirm backup runtime capacity.
- 16. Pre-Installation Check:** Verify UPS provision before installing any IT equipment.
 - Document UPS capacity planning to ensure adequate load headroom for future equipment additions.
- 17. Insurance:** Cover IT assets under comprehensive insurance for financial protection against unforeseen events.
 - Review and renew insurance coverage annually, updating asset values as needed.

Environmental Security Controls

- 18. Data centre Infrastructure:**
 - Ensure air conditioning and fire suppression systems are installed in the data centre.
 - Implement redundant (N+1) cooling units to prevent single points of failure.
- 19. Fire Safety Measures:**
 - Install fire and smoke detectors; regularly check fire extinguishers for pressure, noting last checked and next check dates on cylinders.
 - Conduct periodic fire evacuation drills for staff working in or near the data centre.
- 20. Material Restrictions:**
 - Prohibit toxic and flammable materials from entering Data Centres.
 - Display clear signage listing prohibited materials at data centre entry points.
- 21. Temperature and Humidity Monitoring:**
 - Monitor and control temperature and humidity levels in Data Centres periodically.
 - Maintain logs of temperature and humidity readings for trend analysis and audits.
- 22. Maintenance and Records:**
 - Schedule periodic inspection, testing, and maintenance of air conditioning and fire suppression systems, maintaining records for audit purposes.
 - Assign a responsible owner for tracking and closing out maintenance findings.

Remote Access Connectivity

- 23. Approved Access Application:** They use approved remote access applications to connect to the organization's network from remote locations.
 - Maintain a list of approved remote access tools and block unauthorized applications.
- 24. Approval Process:** Remote access software usage must be approved by the respective functional or department head.
 - Document and retain approval records for audit and compliance purposes.
- 25. Credential Management:** IT department provides user IDs and passwords for all remote access users.

- Enforce periodic password changes and immediate revocation upon employee exit.

Wi-Fi Access Connectivity

- 26. **Location Wi-Fi Provision:** Provide Wi-Fi connectivity across locations/plants for laptop users to access the organization's network.
 - Segregate guest Wi-Fi from the corporate network using a separate SSID and VLAN.
- 27. **Responsibility for Wi-Fi:** IT department is responsible for providing Wi-Fi at meeting rooms, reception, etc., where wired network connectivity is limited.
 - Periodically review Wi-Fi access point placement and signal coverage for gaps.
- 28. **Secure Connectivity:** Ensure all Wi-Fi users connect using secured passwords for network access.
 - Use WPA3 (or strongest available) encryption and rotate Wi-Fi passwords periodically.

Firewall Security Controls

- 29. **Traffic Routing:** All network traffic routed through firewall for incoming and outgoing connections.
 - Block all traffic by default and explicitly allow only required ports and services.
- 30. **Rule Adherence:** Firewalls operate according to business-defined rule sets.
 - Document business justification for each firewall rule for audit traceability.
- 31. **Traffic Management:** Capable of filtering specific traffic types, blocking unauthorized access, and preventing breaches.
 - Enable intrusion detection/prevention (IDS/IPS) features where available.
- 32. **Logging Compliance:** Logs stored as per DOT regulations, aiding audit and investigation.
 - Ensure firewall logs are backed up and retained per regulatory retention requirements.

Router Access Control

- 33. **Physical Security:** Place routers in physically restricted areas accessible only to authorized personnel.
 - Secure routers in locked cabinets or rooms with access logging.
 - Disable unused physical and logical ports on routers to reduce attack surface.